
Aluminum tolerance screening of cayenne pepper (*Capsicum annuum* L.) genotypes at the seedling stage under nutrient solution stress

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Abstract Averaging the concentration inhibited 75% growth (IC₇₅) is shown in the result of orthogonal polynomial regression analysis on plant height, root length, root fresh weight, root dry weight, shoot fresh weight and shoot dry weight yielded an IC₇₅ threshold of ~1600 ppm AlCl₃·6H₂O (≈ 179 ppm Al³⁺ or ≈ 6630 μM Al³⁺), which is proposed as the critical concentration for effective hydroponic screening. The (STI) analysis on the data collected from IC₇₅ hydroponic evaluation on 41 cayenne pepper genotypes provided a robust classification into five tolerance groups, ranging from highly tolerant (HT) to highly sensitive (HS), thereby offering a clear picture of genotypic diversity in response to aluminum stress. The highly tolerant (HT) group was represented by ‘CMK Lolay’ and ‘Tanggo’, which were categorized tolerant across multiple traits. A slightly lower but still strong level of tolerance that was observed in the tolerant (T) group, they were ‘Dumay’, ‘Seulawah Aceh’, ‘Awe Aceh’, ‘Iggo’, ‘Kawat’, ‘Laris’, ‘Perintis’, ‘Landung’, and ‘PBC396’. The rest genotypes were considered as medium tolerant (MT), sensitive (S) and highly sensitive (HS). The HT and T group were potential genetic resources for breeding Al-tolerant cultivars. However, further field validation under acidic soil conditions is recommended.

Keywords: Acid stress, Hot pepper, IC₇₅, Selection, STI

Introduction

Cayenne pepper (*Capsicum annuum* L.) is the most important vegetable crop in Indonesia, contributing significantly to food diversity, income generation, and rural livelihoods. To meet increasing market demand, efforts to expand the planting area are underway into new agricultural frontiers, the acidic soil of Ultisols which cover nearly 40% of the Indonesia’s potentially arable lands

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(Prasetyo and Suriadikarta, 2006). However, crop productivity on these acidic soil is severely constrained by aluminum (Al) toxicity (Shetty *et al.*, 2021). When soil pH drops below 5.0, insoluble aluminosilicates and oxides release Al^{3+} , the phytotoxic ionic form that rapidly inhibits root elongation and division, disrupts membrane stability, interferes with nutrient uptake, and induces oxidative damage (Kocjan *et al.*, 2025). Ultimately, it hampers shoot growth, resulting in stunted plants and reduced yields (Rahman and Upadhyaya, 2021) (Tan *et al.*, 2025).

At the physiological level, there are two complementary strategies to fortify Al tolerance in higher plants. The first is external exclusion, mainly through root exudation of organic acids (malate, citrate, oxalate) that chelate Al^{3+} in the rhizosphere, thereby reducing its activity near root surfaces. The second strategy is internal detoxification, involving Al transport into vacuoles and activation of antioxidant defense pathways (Wei *et al.*, 2021) (Yan *et al.*, 2022).

Root growth inhibition in the apex is the earliest and most consistent symptom of Al toxicity (Shetty *et al.*, 2021), early-stage (seedling) screening in controlled solutions has become a widely used approach for high-throughput evaluation of genotypic differences in Al tolerance (Herison *et al.*, 2022). Hydroponic and nutrient-solution assays allow precise control of Al^{3+} activity and pH, rapid measurement of root-growth parameters (e.g., relative root length, root re-growth), and the integration of physiological and biochemical measurements and tissue Al accumulation (Brhane *et al.*, 2018) (Kulkarni *et al.*, 2021). When appropriately designed to reflect agronomic exposure levels, seedling assays can provide predictive phenotypes for breeding selection (Herison *et al.*, 2022) (Langrai *et al.*, 2025).

Mitigation strategies such as liming and organic amendments (e.g., biochar) have been applied to alleviate soil acidity and improve crop productivity (Zhang *et al.*, 2023); (Dhandapani *et al.*, 2024). However, their effectiveness is inconsistent and often not cost-effective at scale. Lime application can enhance soil physical, chemical and biological properties, yet its impact on lime composition, timing and soil moisture, and requires repeated application over time (Ameyu, 2019). Similarly, biochar can raise soil pH and adsorb Al ions, but its effectiveness dependent on feedstock type and soil characteristic (Shetty *et al.*, 2021). Consequently, breeding and selecting genetically Al-tolerant cultivars offers a more sustainable solution for long-term productivity on acid soils (Wei *et al.*, 2021). Robust selection protocols that exploit plant genetic diversity are critical in identify acid-soil-tolerant genotypes to develop of high-yielding, acid-tolerant varieties. Equally important is the availability tolerant parental lines that can serve as sources of aluminium tolerance controlling gene(s).

Previous studies have shown that pepper seedlings exhibit genotypic variation in shoot and root growth under Al stress (Herison *et al.*, 2022). Since aluminium tolerance in higher plants is quantitatively inherited (Njobvu, 2020) (Chauhan *et al.*, 2021), a broader range of parental lines required for the development of tolerant cultivars.

Accordingly, this study evaluated diverse cayenne pepper genotypes at the seedling stage under controlled hydroponic aluminum exposure, with the objectives of determining the threshold concentration of $\text{AlCl}_3 \cdot 6\text{H}_2\text{O}$ for an effective screening protocol and identifying the most tolerant genotypes as potential sources of aluminum tolerance gene(s).

Materials and methods

The research was conducted at the Greenhouse of the Agroecotechnology Laboratory, Faculty of Agriculture, University of Bengkulu, from January to July 2025, at an altitude of ± 10 meters above sea level. The study was carried out in two stages. The first stage focused on determining the concentration threshold of aluminum chloride ($\text{AlCl}_3 \cdot 6\text{H}_2\text{O}$), and the second stage aimed to evaluate 41 cayenne pepper genotypes for aluminum stress tolerance.

Determination of $\text{AlCl}_3 \cdot 6\text{H}_2\text{O}$ concentration that inhibit 75% growth (IC_{75})

Aluminum stress assay was conducted in a deep water cultivated (DWC) hydroponic system. The planting medium consisted of an AB mix nutrient solution supplemented with $\text{AlCl}_3 \cdot 6\text{H}_2\text{O}$, and placed in rectangular containers measuring 30 x 50 x 20 cm. The experiment used five levels of $\text{AlCl}_3 \cdot 6\text{H}_2\text{O}$ concentrations, i.e. 0 ppm (control), 500 ppm, 1000 ppm, 1500 ppm, and 2000 ppm. Fifteen plants of a randomly selected genotype were used per treatment. Observed variables included plant height, root length, fresh root weight, dry root weight, fresh shoot weight, and dry shoot weight. The collected data were analyzed using analysis of variance (ANOVA) and followed by the orthogonal polynomial at $\alpha=5\%$ to determine the concentration of $\text{AlCl}_3 \cdot 6\text{H}_2\text{O}$ that inhibits growth by 75% (IC_{75}).

Evaluation for aluminum stress tolerance

Evaluation for Al tolerance was conducted in an assay technique similar to the former stage. Forty one cayenne pepper genotypes were evaluated for aluminum stress tolerance, including 'Mada', 'Baja F1', 'Lotanbar', 'Rotane', 'Locker', 'Kopay', 'Radha', 'Ateng', 'Paten', 'Dumay', 'Lokal Legum', 'CMK

Lolay', 'Indrapura Reborn', 'Vitra OP', 'CMK Rejong Aceh', 'Seulawah Aceh', 'Klope Mega Aceh', 'Awe Aceh', 'CMK Tavi', 'Simpatik 17', 'Hellboy', 'Sempurna', 'Horison', 'Anies IPB', 'Iggo', 'Andalas', 'Kawat', 'Gloria', 'Krosa', 'Bali 77', 'Lajang', 'Tenggo', 'Laris', 'Perintis', 'Landung', 'Caman', 'Romario', 'Aka', 'Vitra Unggul' and 'PBC396'. Eighteen plants were tested per genotype. Two $\text{AlCl}_3 \cdot 6\text{H}_2\text{O}$ concentrations were used i.e. 0 ppm (control) and the threshold concentration obtained in the first stage. In control, no aluminum was added to the medium, the pH was maintained between 6.0 and 6.5. For the stress treatment, $\text{AlCl}_3 \cdot 6\text{H}_2\text{O}$ was added to the growing medium, and the pH was maintained at 4.0–4.5 following the method of (Herison *et al.*, 2020). The pH was increased using KOH or decreased using HNO_3 , as needed. The pH solution was measured every three days, and the nutrient solution was replaced every week. Variables measured included plant height, leaf area, number of leaves, leaf greenness, root length, root volume, root fresh weight, root dry weight, shoot fresh weight, and shoot dry weight. Data analysis was conducted by ANOVA and means clustering by the Scott-Knott test at $\alpha=5\%$.

Stress tolerance index (STI) and genotype classification

The stress tolerance index (STI) was calculated using the formula which was used by (Solangi *et al.*, 2021):

$$\text{STI} = \frac{(Y_p \times Y_s)}{(\bar{Y}_p)^2}$$

where STI, Y_p , Y_s and $\bar{Y}_p$ represents the stress tolerance index, the observed value under non-stress conditions (control), the observed value under stress conditions, and the average value under non-stress condition over all genotypes evaluated. The average of tolerance index over all variables was classified into five proportional tolerance levels: highly tolerant (HT), tolerant (T), moderately tolerant (MT), sensitive (S), and highly sensitive (HS).

Results

Determination of aluminum concentration for selection

The growth pattern observed under varying aluminum concentrations indicated a clear reduction in plant height as the concentration of $\text{AlCl}_3 \cdot 6\text{H}_2\text{O}$ increased. In the control treatment (without $\text{AlCl}_3 \cdot 6\text{H}_2\text{O}$), plant height increased by approximately 8 to 10 cm weekly. In the treatment of 500 ppm, plant height increased by 3 to 10 cm from the previous week. For the 1000 ppm, growth was slower, with a height increase of 2 to 3 cm weekly. The treatment of 1500 ppm resulted in a minimal growth increased of 0.45 to 1.66 cm per week, while in

2000 ppm, plant height increased by only 0.17 to 1.04 cm during weeks 2 and 3, and began to stagnant, indicating that the plants were unable to survive under severe aluminum stress (Figure 1).

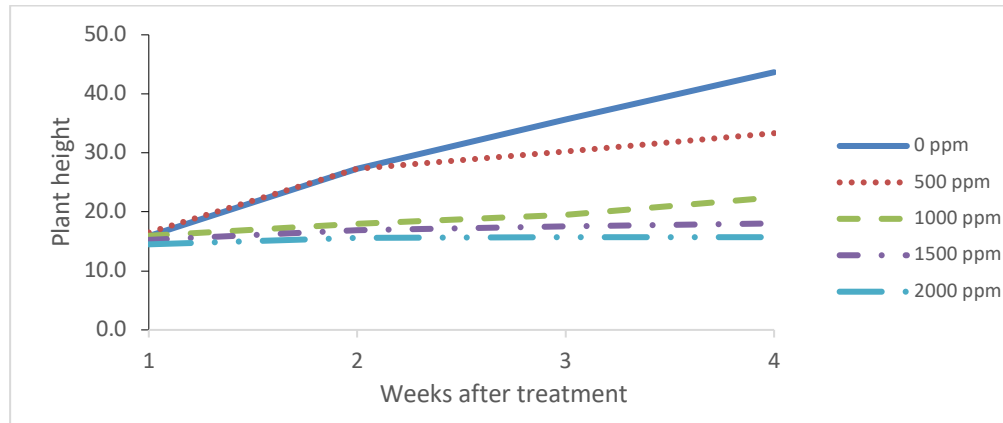


Figure 1. Growth pattern of cayenne pepper plants during determination of the threshold aluminium concentration

Based on the results of the variance analysis, the treatments of varying $\text{AlCl}_3 \cdot 6\text{H}_2\text{O}$ concentrations significantly affected all growth parameters, including plant height, leaf greenness, root length, root volume, fresh root weight, dry root weight, fresh shoot weight, and dry shoot weight. The coefficient of variation (CV) values ranged from 5.1% to 16.9%, indicating a high level of accuracy in the experiment. The coefficient of variation is used to assess the accuracy of treatments, with lower CV values representing higher accuracy (Table 1).

Table 1. Summary of the ANOVA in determination the threshold aluminum concentration

Variable	F Value	CV (%)
Plant height	304.083*	10.4
Root length	22.054*	9.2
Root fresh weight	338.676*	7.4
Root dry weight	64.963*	16.9
Shoot fresh weight	307.758*	8.3
Shoot dry weight	263.715*	8.7

Note: * indicates significant at the 5% level, ns = not significant.

Regression analyses showed distinct responses of seedling growth traits to $\text{AlCl}_3 \cdot 6\text{H}_2\text{O}$ stress (Table 2). The plant height curve fitted the quadratic equation of $y = 7\text{E}^{-06}\text{x}^2 - 0.0279\text{x} + 27.98$ with a high coefficient of determination ($\text{R}^2 =$

0.987), which was the most sensitive trait, showing 75% inhibition at only 1006 ppm. On the contrary, the root length responses followed a linear decline ($y = -0.0034x + 14.056$, $R^2 = 0.891$) and required as high as 3101 ppm before being reduced to the same threshold, which was categorized comparatively higher tolerance.

Table 2. Summary of the results of the Orthogonal Polynomial at $\alpha = 5\%$

Variable	Orthogonal Polynomial Equation	IC ₇₅ (ppm)*
Plant height	$y = 7E-06x^2 - 0.0279x + 27.98$ ($R^2 = 0.987$)	1006
Root length	$y = -0.0034x + 14.056$ ($R^2 = 0.8911$)	3100
Root fresh weight	$y = 4E-07x^2 - 0.0017x + 2.1614$ ($R^2 = 0.9629$)	1445
Root dry weight	$y = 5E-08x^2 - 0.0002x + 0.2757$ ($R^2 = 0.9572$)	1626
Shoot fresh weight	$y = 3E-06x^2 - 0.0102x + 11.297$ ($R^2 = 0.9868$)	1444
Shoot dry weight	$y = 4E-07x^2 - 0.0016x + 1.811$ ($R^2 = 0.9645$)	1223

Note: *Threshold concentration of $AlCl_3 \cdot 6H_2O$ which inhibits 75% growth.

Biomass-related traits showed intermediate sensitivities. Root fresh weight responses followed the trend equation of $y = 4E-07x^2 - 0.0017x + 2.1614$ ($R^2 = 0.963$), and shoot fresh weight, described by $y = 3E-06x^2 - 0.0102x + 11.297$ ($R^2 = 0.987$), both reached 75% inhibition at ~1444 ppm. Root dry weight followed $y = 5E-08x^2 - 0.0002x + 0.2757$ ($R^2 = 0.957$), with growth reduced at 1626 ppm, while shoot dry weight fitted the equation of $y = 4E-07x^2 - 0.0016x + 1.811$ ($R^2 = 0.965$), was 75% inhibited at 1223 ppm. The consistently high R^2 values (>0.95 for most traits) indicate that the regression models accurately captured the close relationship between aluminum concentration and growth inhibition.

These results indicated that plant height and shoot dry weight were more reliable early indicators of aluminum stress, as their low inhibition thresholds and strong model fits. Conversely, root length was the least sensitive to Al stress, with a slightly weaker model fit ($R^2 = 0.891$), suggesting that root elongation still occur under stress, and it may be influenced by other uncontrolled factors. Averaging across all traits, the inhibitory threshold of ~1641 ppm $AlCl_3 \cdot 6H_2O$ (equal to 183 ppm Al^{3+} or 6790 μM Al^{3+}) is proposed as the critical concentration for effective hydroponic screening techniques. This result provides a practical reference for distinguishing tolerant from sensitive genotypes in future cayenne pepper breeding programs aimed at improving acidic soil adaptation

Plant performance under aluminum stress

The results of analysis of variance showed that under aluminium stress of 1600 ppm $AlCl_3 \cdot 6H_2O$, there were significantly variation among genotypes in plant height, leaf greenness, leaf area, number of leaves, root length, root volume, fresh root weight, dry root weight, fresh shoot weight, and dry shoot weight. The

coefficient of variation (CV) indicated the accuracy of the treatment. The CV values ranged from 3.07% to 39.37%, suggesting that the research accuracy was relatively high (Table 3).

Table 3. Summary of the analysis of variance (ANOVA)

Variables	F calculated	CV
Plant height	22.051*	9.16%
Leaf greenness	107.210*	3.07%
Number of leaves	7.919*	7.25%
Leaf area	5.036*	39.37%
Root length	117.279*	4.58%
Root Volume	34.996*	15.40%
Root fresh weight	169.966*	7.74%
Root dry weight	140.720*	10.75%
Shoot fresh weight	636.957*	5.11%
Shoot dry weight	658.843*	6.02%

Note *: significantly different at $\alpha = 5\%$

The Scott–Knott grouping revealed different traits resulted in different clusters. For plant height, the tallest group consisted of 'Laris', 'Vitra Unggul', 'Horison', and 'Lokal Legum', with values ranging from 12.68 to 14.06 cm. The intermediate group, including 'CMK Lolay', 'Simpatik 17', 'Iggo', 'Labek', and 'PBC396', ranged between 10.20 and 11.63 cm. while the majority of other genotypes, such as 'Mada', 'Baja F1', and 'Rotane', fell into the lowest group with heights from 6.21 to 9.39 cm (Table 4).

In terms of leaf greenness, the superior cluster was formed by 'Tanggo', 'Caman', and 'Kawat', showing the highest SPAD values between 41.51 and 44.74. Conversely, the least green genotypes, 'Bali 77', 'CMK Tavi', and 'Simpatik 17', clustered in the lowest group (SPAD 21.01 to 23.33). A broad intermediate range (SPAD 24.56 to 39.73) included genotypes such as 'Radha', 'Andalas', 'Romario', and 'Vitra OP'.

For leaf number, the superior cluster comprised 'Simpatik 17', 'Horison', 'Lajang', 'Labek', and 'Laris', with 8.22 to 9.11 leaves per plant. By contrast, 'Baja F1', 'Locker', 'Radha', and 'Ateng' belonged to the lowest group, with only 5.67 to 5.89 leaves, while the majority of genotypes, such as 'Mada', 'Rotane', 'CMK Rencong Aceh', and 'Hellboy', occupied an intermediate group (around 6.11 to 7.78 leaves). The analysis of leaf area further differentiated the accessions. The largest leaf areas were recorded in 'Simpatik 17', 'Horison', 'Iggo', 'Labek', and 'Laris', with values between 0.67 and 1.00 cm². The lowest-performing group included 'Sempurna', 'Klope Mega Aceh', 'Anies IPB', and 'Aka', all with areas below 0.50 cm². Intermediate performers, such as 'CMK Rencong Aceh', 'Hellboy', and 'Locker', showed leaf areas ranging 1.30 to 1.96 cm² (Table 4).

Table 4. Plant height, leaf greenness, number of leaves, leaf area and root length of 41 genotypes under aluminum stress

Genotypes	Plant height (cm)	Leaf greenness	Number of leaves	Leaf Area (cm ²)	Root length (cm)
'Mada'	7.37 d	27.88 g	6.44 c	1.34 b	8.78 f
'Baja F1'	6.66 d	29.88 f	5.89 d	1.84 a	10.76 d
'Indarapura Reborn'	6.21 d	24.87 i	6.11 d	0.68 c	5.67 i
'Rotane'	7.02 d	26.57 h	6.78 c	1.44 b	10.42 d
'Locker'	6.70 d	28.90 g	5.67 d	1.30 b	14.66 a
'Kopay'	8.70 c	24.56 i	6.44 c	0.62 c	9.53 e
'Radha'	6.34 d	32.67 e	5.78 d	0.70 c	9.30 e
'Ateng'	6.38 d	30.37 f	6.11 d	0.68 c	5.17 i
'Paten'	6.71 d	31.19 f	6.56 c	0.62 c	5.11 i
'Dumay'	8.44 c	31.80 f	7.78 b	0.50 c	6.73 h
'Lokal Legum'	12.68 a	31.20 f	8.00 b	0.60 c	13.56 b
'CMK Lolay'	11.63 b	33.12 e	8.11 b	0.53 c	12.43 c
'Lontanbar'	7.19 d	27.79 g	7.78 b	0.69 c	6.44 h
'Vitra Unggul'	13.52 a	26.58 h	7.44 b	0.64 c	12.44 c
'CMK Rencong Aceh'	8.30 c	32.52 e	6.44 c	1.96 a	9.53 e
'Seulawah Aceh'	6.98 d	33.72 e	7.56 b	1.05 b	11.78 c
'Klope Mega Aceh'	9.39 c	28.50 g	7.00 c	0.50 c	6.33 h
'Awe Aceh'	8.13 c	38.66 c	6.67 c	0.60 c	14.09 a
'CMK Tavi'	7.53 d	23.33 j	7.78 b	0.26 c	9.11 e
'Simpatik 17'	10.57 b	22.84 j	8.22 a	0.67 c	7.21 g
'Hellboy'	7.67 d	28.53 g	7.33 b	1.38 b	6.30 h
'Sempurna'	7.50 d	27.88 g	7.89 b	0.21 c	9.04 e
'Horison'	12.99 a	24.98 i	8.89 a	1.04 b	7.67 g
'Anies IPB'	7.94 c	28.19 g	7.89 b	0.47 c	5.33 i
'Iggo'	9.88 b	28.98 g	8.33 a	0.79 c	9.92 e
'Andalas'	6.94 d	38.41 c	8.11 b	0.95 c	11.18 d
'Kawat'	6.71 d	41.51 b	7.33 b	1.35 b	7.08 g
'Gelora'	7.27 d	28.33 g	7.22 c	0.63 c	9.43 e
'Ferosa'	7.10 d	27.61 g	7.00 c	0.45 c	6.92 g
'Bali 77'	8.88 c	21.01 k	7.89 b	0.49 c	8.33 f
'Lajang'	9.04 c	38.52 c	8.22 a	1.38 b	10.92 d
'Tanggo'	7.21 d	44.74 a	7.78 b	1.24 b	14.49 a
'Laris'	14.06 a	35.97 d	9.11 a	1.00 c	7.96 f
'Perintis'	6.81 d	39.73 c	8.56 a	1.07 b	12.96 b
'Landung'	6.91 d	36.11 d	7.89 b	2.23 a	9.20 e
'Labek'	10.73 b	37.27 d	8.44 a	0.74 c	8.53 f
'Caman'	7.66 d	42.19 b	7.89 b	1.45 b	11.97 c
'Romario'	8.23 c	35.52 d	7.89 b	0.68 c	9.60 e
'Aka'	8.08 c	25.88 h	7.11 c	0.47 c	9.22 e
'Vitra OP'	8.64 c	34.58 e	6.78 c	1.35 b	6.86 g
'PBC396'	10.20 b	37.20 d	7.10 c	1.05 b	12.10 c

Note: numbers followed by the same letter are not significantly different according to Scott-Knott at $\alpha = 5\%$.

For root length, the longest roots were observed in 'Locker', 'Awe Aceh', and 'Tanggo', grouped at the top with values between 14.09 and 14.66 cm. The lowest group comprised 'Indrapura Reborn', 'Ateng', 'Paten', and 'Anies IPB', whose root lengths were restricted to 5.11 to 5.67 cm. A wide middle range of 6.73 to 13.56 cm included genotypes such as 'Baja F1', 'Rotane', 'CMK Rencong Aceh', and 'Romario' (Table 4).

The group with the highest root volume, with an average ranging from 0.52 mm³ to 0.54 mm³, included the genotypes 'Vitra Unggul' and 'Landung'. The group with the lowest root volume, with an average ranging from 0.10 mm³ to 0.13 mm³, included the genotypes 'Indrapura Reborn', 'Kopay', 'Paten', 'CMK Lolay', 'Lontanbar', 'Klope Mega Aceh', 'Hellboy', 'Ferosa', 'Bali 77', 'Lajang', 'Vitra OP', and 'Horison' (Table 5).

The group of the highest root fresh weight, with an average ranging from 0.45 g to 0.48 g, included the genotypes 'Aka', 'Labek', 'Caman', and 'Landung'. In contrast, the group with the lowest root fresh weight, ranging from 0.06 g to 0.08 g, included the genotypes 'Indrapura Reborn', 'Lajang', 'Lontanbar', 'Hellboy', 'Klope Mega Aceh', 'Ferosa', 'Bali 77', and 'Kopay'. The group with the highest root dry weight, ranging from 0.13 g to 0.14 g, included the genotypes 'Landung', 'Labek', 'Caman', and 'Aka'. The group with the lowest root dry weight, included the genotypes 'Indrapura Reborn', 'Kopay', 'Lontanbar', 'Klope Mega Aceh', 'Ferosa', 'Bali 77', and 'Lajang' (Table 5).

The group with the highest shoot fresh weight, with an average ranging from 4.92 g to 5.03 g, included the genotypes 'Romario' and 'Dumay'. In contrast, the group with the lowest shoot fresh weight, with an average ranging from 0.41 g to 0.52 g, included the genotypes 'Lajang', 'Klope Mega Aceh', 'Lontanbar', 'Locker', 'Indrapura Reborn', and 'Ferosa'. The highest shoot dry weight group, with an average ranging from 1.76 g to 1.78 g, were the genotypes 'Dumay' and 'Romario'. The group with the lowest shoot dry weight, with an average ranging from 0.06 g to 0.11 g, included the genotypes 'Lajang', 'Ferosa', 'Klope Mega Aceh', 'Vitra Unggul', 'Locker', and 'Horison' (Table 5).

Considering all variables, genotypes exhibited superior growth performance and clustered in the highest groups were 'Laris', 'Vitra Unggul', 'Landung', 'Labek', 'Caman', 'Aka', 'Romario', 'Dumay', 'PBC396', and 'Tanggo'. These genotypes combined vigorous shoot growth (fresh and dry weight) with strong root development under stress condition. The moderately performing genotypes showing intermediate performance were 'Iggo', 'CMK Rencong Aceh', 'Perintis', 'Simpatik 17', 'Seulawah Aceh', 'Gelora', 'Andalas', and 'Radha'. The rest genotypes were group into the lowest performing group under stress condition.

Table 5. Root volume, root fresh and dry weight, and shoot fresh and dry weight of 41 genotypes under aluminum stress

Genotype	Root volume (mm ³)	Root fresh weight (g)	Root dry weight (g)	Shoot fresh weight (g)	Shoot dry weight (g)
'Mada'	0.18 f	0.11 j	0.03 h	1.48 i	0.50 j
'Baja F1'	0.38 c	0.25 f	0.08 e	1.46 i	0.42 k
'Indarapura Reborn'	0.10 g	0.06 k	0.01 j	0.53 n	0.15 n
'Rotane'	0.26 e	0.14 i	0.04 g	1.12 k	0.22 m
'Locker'	0.40 c	0.22 g	0.05 f	0.48 n	0.08 o
'Kopay'	0.10 g	0.08 k	0.01 j	0.66 m	0.14 n
'Radha'	0.33 d	0.19 h	0.04 g	1.92 g	0.61 i
'Ateng'	0.23 e	0.15 i	0.03 h	0.85 l	0.21 n
'Paten'	0.10 g	0.11 j	0.02 i	0.67 m	0.16 n
'Dumay'	0.36 c	0.37 b	0.10 c	5.03 a	1.78 a
'Lokal Legum'	0.32 d	0.26 e	0.07 e	0.93 l	0.27 m
'CMK Lolay'	0.13 g	0.11 j	0.03 h	0.67 m	0.17 n
'Lontanbar'	0.11 g	0.06 k	0.01 j	0.51 n	0.13 o
'Vitra Unggul'	0.52 a	0.37 b	0.12 b	2.25 f	0.66 h
'CMK Rencong Aceh'	0.48 b	0.26 e	0.06 f	2.79 e	0.85 g
'Seulawah Aceh'	0.23 e	0.23 f	0.04 g	1.93 g	0.53 J
'Klope Mega Aceh'	0.10 g	0.07 k	0.01 j	0.51 n	0.11 o
'Awe Aceh'	0.39 c	0.24 f	0.06 f	1.58 i	0.48 j
'CMK Tavi'	0.18 f	0.12 j	0.02 i	0.67 m	0.18 n
'Simpatik 17'	0.30 d	0.21 g	0.04 g	1.09 k	0.30 l
'Hellboy'	0.10 g	0.07 k	0.01 j	0.67 m	0.16 n
'Sempurna'	0.30 d	0.23 f	0.07 f	1.03 k	0.30 l
'Horison'	0.13 g	0.13 i	0.02 i	0.60 m	0.11 o
'Anies IPB'	0.17 f	0.12 j	0.02 i	1.16 k	0.32 l
'Iggo'	0.43 b	0.26 e	0.06 f	1.42 i	0.42 k
'Andalas'	0.22 e	0.18 h	0.03 h	1.26 j	0.35 l
'Kawat'	0.22 e	0.19 h	0.03 h	2.90 e	0.92 f
'Gelora'	0.27 e	0.21 g	0.04 g	1.49 i	0.45 k
'Ferosa'	0.10 g	0.07 k	0.01 j	0.52 n	0.08 o
'Bali 77'	0.11 g	0.07 k	0.01 j	0.71 m	0.17 n
'Lajang'	0.10 g	0.06 k	0.01 j	0.41 n	0.06 o
'Tanggo'	0.49 b	0.34 c	0.07 e	3.48 c	1.29 c
'Laris'	0.18 f	0.22 g	0.05 g	1.78 h	0.48 j
'Perintis'	0.34 d	0.31 d	0.09 d	2.16 f	0.66 h
'Landung'	0.54 a	0.48 a	0.13 a	3.44 c	1.27 c
'Labek'	0.44 b	0.46 a	0.13 a	3.16 d	1.12 e
'Caman'	0.39 c	0.47 a	0.13 a	3.39 c	1.23 d
'Romario'	0.31 d	0.38 b	0.11 b	4.92 a	1.76 a
'Aka'	0.19 f	0.45 a	0.14 a	4.39 b	1.39 b
'Vitra OP'	0.12 g	0.21 g	0.05 g	2.09 f	0.60 I
'PBC396'	0.43 b	0.37 b	0.12 b	4.40 b	1.29 c

Note: numbers followed by the same letter are not significantly different according to Scott Knott at $\alpha = 5\%$.

Stress tolerance index (STI) and genotype classification

Examination of STI values across individual traits showed that tolerance was not uniformly distributed across growth components. For plant height, several genotypes exhibiting high STI, 'Vitra Unggul' (0.57), 'Lokal Legum' (0.55), 'Laris' (0.52) and 'Horison' (0.51), sustained vertical growth under stress, whereas 'Andalas' (0.19) and 'Paten' (0.23) were highly inhibited. Leaf greenness (LG) became one of the most informative indicators, with exceptionally high STI values in 'Awe Aceh' (0.93) and 'Kawat' (0.95). Conversely, 'CMK Tavi' (0.37) and 'Bali 77' (0.37) were more highly reduced. For leaf area (LA), 'Seulawah Aceh' (1.01) and 'Awe Aceh' (0.99) exhibited highly tolerance, whereas 'Mada' (0.38) and 'Ateng' (0.27) were among the most sensitive (Table 6).

The root traits showed the greatest variation, which reflect their central role in aluminum stress adaptation. For root length, genotype 'CMK Lolay', 'Seulawah Aceh' and 'Awe Aceh' consistently showed higher STI compared other genotypes. In contrast, 'Indarapura Reborn' and 'Lontanbar' exhibited low STI values across root traits, consistent with their HS classification. For the shoot-related traits (SFW, SDW), 'Dumay' (SFW = 0.55; SDW = 0.47) and 'Romario' (SDW = 0.32) demonstrated relatively stronger shoot biomass accumulation despite stress. Conversely, 'Mada' and 'Indarapura Reborn' recorded the lowest STI values, reaffirming their high susceptibility (Table 6).

Considering all variables observed, the STI analysis classified the evaluated cayenne pepper genotypes into five tolerance groups, they were highly tolerant (HT), tolerant (T), medium tolerant (MT), sensitive (S), and highly sensitive (HS). The classification indicated genotypic diversity in response to aluminum stress. The highly tolerant (HT) group was represented by 'CMK Lolay' and 'Tanggo', with average STI values above 0.40. The STI value in the tolerant (T) group, which included 'Dumay', 'Seulawah Aceh', 'Awe Aceh', 'Iggo', 'Kawat', 'Laris', 'Perintis', 'Landung', and 'PBC396', was a slightly lower, ranging between 0.32 and 0.37. The medium-tolerant (MT) group was relatively large and diverse, consisting of 'Locker', 'Radha', 'Lokal Legum', 'Vitra Unggul', 'CMK Rencong Aceh', 'Anies IPB', 'Andalas', 'Labek', 'Caman', 'Romario', and 'Aka'. These genotypes showed STI values between 0.27 and 0.32. In contrast, the sensitive (S) group included 'Baja F1', 'Klope Mega Aceh', 'CMK Tavi', 'Simpatik 17', 'Hellboy', 'Horison', 'Gloria', 'Ferosa', 'Bali 77', 'Lajang', and 'Vitra OP'. These genotypes displayed STI values between 0.23 and 0.27. Finally, the highly sensitive (HS) group consisted of 'Mada', 'Indarapura Reborn', 'Rotane', 'Ateng', 'Paten', 'Lontanbar', and 'Sempurna', all of which recorded STI values ≤ 0.23 (Table 6). These genotypes presented severe reductions both root and shoot traits.

Table 6. Stress tolerance index (STI) value and tolerance classification of 41 cayenne pepper genotypes under aluminium stress of 1600 ppm AlCl₃.6H₂O

Genotype	STI										Average STI	Class
	PH	LG	NL	LA	RL	RV	RF W	RD W	SF W	SD W		
'Mada'	0.24	0.41	0.56	0.19	0.38	0.02	0.03	0.05	0.09	0.09	0.20	HS
'Baja F1'	0.24	0.56	0.48	0.26	0.56	0.07	0.07	0.15	0.06	0.07	0.25	S
'Indarapura Reborn'	0.22	0.48	0.54	0.07	0.29	0.02	0.02	0.02	0.05	0.04	0.17	HS
'Rotane'	0.23	0.51	0.52	0.16	0.54	0.04	0.04	0.08	0.05	0.04	0.22	HS
'Locker'	0.22	0.58	0.45	0.26	0.69	0.22	0.12	0.20	0.03	0.02	0.28	MT
'Kopay'	0.33	0.39	0.52	0.14	0.57	0.09	0.05	0.04	0.05	0.08	0.22	HS
'Radha'	0.22	0.53	0.46	0.09	0.79	0.28	0.13	0.18	0.16	0.23	0.30	MT
'Ateng'	0.21	0.50	0.54	0.16	0.27	0.08	0.06	0.08	0.05	0.05	0.20	HS
'Paten'	0.23	0.74	0.52	0.10	0.25	0.04	0.04	0.05	0.04	0.04	0.20	HS
'Dumay'	0.32	0.77	0.68	0.05	0.33	0.07	0.12	0.21	0.55	0.47	0.35	T
'Lokal Legum'	0.55	0.75	0.73	0.10	0.66	0.06	0.07	0.12	0.06	0.07	0.31	MT
'CMK Loday'	0.48	0.80	0.70	0.08	1.46	0.14	0.09	0.14	0.11	0.05	0.40	HT
'Lontanbar'	0.23	0.49	0.65	0.13	0.33	0.04	0.02	0.02	0.04	0.03	0.20	HS
'Vitra Unggul'	0.57	0.42	0.63	0.11	0.61	0.08	0.10	0.18	0.08	0.24	0.30	MT
'CMK Rencong Aceh'	0.27	0.62	0.53	0.27	0.68	0.09	0.08	0.13	0.22	0.19	0.31	MT
'Seulawah Aceh'	0.24	0.59	0.61	0.14	1.01	0.15	0.16	0.17	0.14	0.13	0.33	T
'Klope Mega Aceh'	0.38	0.68	0.62	0.10	0.56	0.09	0.06	0.05	0.08	0.03	0.26	S
'Awe Aceh'	0.25	0.93	0.57	0.10	0.99	0.19	0.11	0.19	0.10	0.12	0.35	T
'CMK Tavi'	0.26	0.37	0.66	0.06	0.55	0.16	0.10	0.09	0.06	0.05	0.23	S
'Simpatik 17'	0.41	0.43	0.70	0.14	0.37	0.04	0.08	0.13	0.07	0.08	0.24	S
'Hellboy'	0.24	0.48	0.64	0.20	0.68	0.07	0.07	0.06	0.09	0.04	0.25	S
'Sempurna'	0.22	0.47	0.62	0.07	0.53	0.04	0.06	0.09	0.07	0.06	0.22	HS
'Horison'	0.51	0.56	0.80	0.19	0.47	0.02	0.05	0.06	0.03	0.03	0.27	S
'Anies IPB'	0.30	0.63	0.70	0.06	0.61	0.17	0.11	0.10	0.13	0.09	0.29	MT
'Iggo'	0.36	0.68	0.71	0.12	0.72	0.27	0.21	0.24	0.17	0.12	0.35	T
'Andalas'	0.19	0.89	0.70	0.12	0.63	0.04	0.05	0.05	0.14	0.10	0.29	MT
'Kawat'	0.20	0.95	0.62	0.17	0.55	0.16	0.16	0.13	0.23	0.22	0.34	T
'Gelora'	0.24	0.73	0.70	0.09	0.44	0.05	0.06	0.07	0.13	0.12	0.26	S
'Ferosa'	0.24	0.67	0.62	0.07	0.59	0.03	0.02	0.03	0.04	0.02	0.23	S
'Bali 77'	0.28	0.37	0.67	0.08	0.65	0.06	0.04	0.04	0.06	0.04	0.23	S
'Lajang'	0.30	0.74	0.73	0.19	0.52	0.03	0.03	0.03	0.03	0.01	0.26	S
'Tanggo'	0.23	0.93	0.60	0.20	0.89	0.31	0.22	0.29	0.25	0.32	0.42	HT
'Laris'	0.52	0.87	0.82	0.13	0.28	0.17	0.21	0.25	0.20	0.14	0.36	T
'Perintis'	0.20	0.76	0.67	0.13	0.83	0.19	0.16	0.34	0.23	0.16	0.36	T
'Landung'	0.21	0.79	0.65	0.25	0.45	0.16	0.20	0.36	0.15	0.19	0.34	T
'Labek'	0.31	0.72	0.76	0.17	0.39	0.07	0.10	0.10	0.11	0.13	0.28	MT
'Caman'	0.24	0.89	0.52	0.28	0.58	0.07	0.13	0.16	0.11	0.16	0.31	MT
'Romario'	0.25	0.56	0.51	0.16	0.55	0.11	0.23	0.30	0.22	0.32	0.32	MT
'Aka'	0.25	0.62	0.47	0.06	0.45	0.06	0.23	0.40	0.21	0.29	0.30	MT
'Vitra OP'	0.26	0.58	0.46	0.19	0.46	0.02	0.06	0.10	0.14	0.14	0.24	S
'PBC396'	0.29	0.62	0.59	0.18	0.34	0.07	0.08	0.07	0.16	0.14	0.36	T

Note: PH = plant height, LG = leaf greenness, NL = number of leaves, LA = leaf area, RL = root length, RV = root volume, RFW = root fresh weight, RDW = root dry weight, SFW = shoot fresh weight, SDW = shoot dry weight. Tolerance classification: STI ≤ 0.23 (highly sensitive, HS), 0.23 < STI ≤ 0.27 (sensitive, S), 0.27 < STI ≤ 0.32 (medium tolerant, MT), 0.32 < STI ≤ 0.37 (tolerant, T), STI > 0.37 (highly tolerant, HT)

Discussion

IC₇₅ determination

The present hydroponic assay showed trait-specific responses of cayenne pepper seedlings to increasing $\text{AlCl}_3 \cdot 6\text{H}_2\text{O}$ concentrations. Plant height was most sensitive to aluminum stress, where a 75% reduction relative to control was obtained at ~1006 ppm. In contrast, root length was more tolerant, requiring ~3101 ppm to reach the same level of inhibition. Biomass traits (fresh and dry weights of shoots and roots) showed intermediate thresholds (~1,200–1,600 ppm). These findings implied the differential sensitivity of shoot and root traits to aluminum stress and signified the importance of multi-trait evaluation for effective screening. Such patterns are consistent with other crops, where aluminum toxicity produces complex, trait-dependent responses that can be exploited in breeding programs (Herison *et al.*, 2022; Ofoe *et al.*, 2023).

The high coefficients of determination (R^2) for most equation models (≥ 0.95 for plant height, shoot biomass, and root biomass) indicated that the models explained nearly all variation across the concentration range, which provided strong confidence in the estimated inhibitory concentrations obtained. For example, the R^2 of 0.987 for plant height means that ~98.7% of the observed variation was represented by the model, which means that it was a robust predictor. By contrast, the lower R^2 for root length (0.891) suggests that additional environmental factors may influence this trait. This aligns with findings that root elongation is often more variable under Al stress, due to genotype-specific root system architecture and physiological plasticity (Wei *et al.*, 2021).

Our calculated average inhibitory concentration (IC_{75}) across traits was ~1641 ppm $\text{AlCl}_3 \cdot 6\text{H}_2\text{O}$. This averaged threshold provided sufficient stress to differentiate tolerant and sensitive genotypes without completely suppressing growth in hydroponic screening. Comparable studies in finger millet identified ~100 μM Al^{3+} as the optimal concentration for genotype discrimination (Brhane *et al.*, 2018). Thus, ~1600 ppm is recommended as an effective threshold for cayenne pepper screening in deep water culture hydroponics.

Mechanistically, aluminum toxicity interferes with root cell wall expansion, triggers oxidative stress, disrupts nutrient uptake, and ultimately restricts both root and shoot development. Tolerant genotypes mitigate these effects through mechanisms such as organic acid exudation, vacuolar sequestration, and regulation by transcription factors (Wei *et al.*, 2021) (Ofoe *et al.*, 2023).

al., 2023). These mechanisms help explain why shoot biomass collapsed earlier while root elongation persisted longer in some cases. Importantly, because aluminum tolerance is quantitatively inherited and controlled by multiple genes, identifying diverse tolerant parents is critical in breeding programs (Herison *et al.*, 2022).

Genotype differences under Al stress condition

The genotypes evaluated exhibited different responses to the applied aluminum concentrations. Genotypes under aluminum stress expressed a plant height inhibition compared to non-stressed genotypes (Musyimi, 2023). Plants subjected to aluminum stress typically showed smaller and stunted shoots. Aluminum stress can cause a significant reduction in shoot dry weight, plant leaves to yellow and drop prematurely, reducing the leaf surface area available for photosynthesis. Exposure to aluminum also leads to smaller and thinner leaf blades, which is associated with reduced cell size and a lower number of stomata (Rahman *et al.*, 2024).

When dissolved aluminum enters the root zone, one of the primary effects is the inhibition of root growth. Aluminum can damage the root system by inhibiting root elongation, reducing lateral root growth, and causing deformations such as root tip thickening. This inhibition is caused by disruptions in cell division and elongation processes in the root meristem. Additionally, aluminum can interact with the cell walls of roots, increasing cell wall rigidity, which limits cell expansion and reduces cell wall elasticity, preventing normal root growth and reduce the ability of roots in absorbing water and nutrients from the soil (Fan *et al.*, 2020). Some genotypes showed relatively good growth despite being grown under the same stress conditions. This differences is attributed to the genetic factors of each plant (Njobvu, 2020) (Devi *et al.*, 2021) (Maksimović *et al.*, 2020).

STI and genotype classification

Leaf greenness (LG) acted as a particularly discriminating trait. This trait frequently associated with improved stress resilience and stability (Ofoe *et al.*, 2023). Genotypes 'Awe Aceh' and 'Kawat' showed high STI values for LG. They might preserved chlorophyll content under aluminum stress. Conversely, several HS genotypes lost chlorophyll rapidly. Chlorosis is one of the common

symptoms of Al-induced damage to photosynthetic apparatus observed across species (Yan *et al.*, 2022).

Root traits (root length, root volume, root fresh and dry weight) were major drivers of the HT/T vs HS/S separation as they showed high spread of STI values. Tolerant genotypes-maintained root biomass and architecture under stress, whereas sensitive genotypes showed drastic root suppression. This pattern is in harmony with numerous screening studies and physiological reviews that root morphology and biomass retention under Al exposure are strong predictors of tolerance (Bhukya *et al.*, 2024)). Root integrity and continued growth under Al stress allow sustained water and nutrient uptake, which in turn supports shoot growth and yield components.

Shoot biomass (SFW, SDW) also contributed to the classification. Genotypes such as 'Dumay' and 'Romario' revealed relatively high STI in shoot traits with good root performance. This emphasizes that effective tolerance is often multi-component. Maintaining root function alone is necessary but not always sufficient. The genotype must also support shoot maintenance to translate tolerance into productive potential (Dhandapani *et al.*, 2024).

The STI-based classification indicates the diversity of adaptive responses to aluminum stress among cayenne pepper genotypes. While highly tolerant genotypes such as 'CMK Lolay' and 'Tanggo' showed broad resilience across traits, others exhibited trait-specific tolerance, either in leaf greenness, shoot biomass, or root development. This differential expression of tolerance mechanisms suggests that pyramiding complementary traits from diverse genotypes could be a promising strategy in breeding aluminum-tolerant cultivars adapted to acidic soils.

The STI classification provides a concise, comparative view of how the evaluated capsicum genotypes withstand aluminum stress. 'CMK Lolay' and 'Tanggo' were uniquely classified as highly tolerant (HT), while a larger set (e.g., 'Dumay', 'Seulawah Aceh', 'Awe Aceh', 'Iggo', 'Kawat', 'Laris', 'Perintis', 'Landung', 'PBC396') formed the tolerant (T) group. These genotypes consistently expressed resilience across multiple traits. They may possess both morphological and physiological mechanisms conferring aluminum tolerance. Medium-tolerant (MT) and sensitive (S) genotypes filled the middle classes, and several genotypes were assigned to the highly sensitive (HS) class. This stratification indicates genetic variation for aluminum response across the trial population and provides a practical guide for selecting parental material for breeding programs.

From a breeding and screening perspective the STI results lead to three practical recommendations. First, prioritize 'CMK Lolay' and 'Tanggo' (HT) as donor parents in crossing programs because they show high resilience. Second, include several T genotypes (e.g., 'Dumay', 'Seulawah Aceh', 'Landung', 'Perintis', 'PBC396') to capture complementary tolerance mechanisms and allelic diversity. Third, use HS genotypes as contrasting parents in mapping populations studies as their sensitivity helps revealing tolerance loci (Chauhan *et al.*, 2021).

Tolerant genotypes exhibit similar characteristics under both stressed and non-stressed conditions. The mechanism of tolerance to aluminum toxicity is thought to operate externally, where aluminum detoxification occurs through the secretion of oxalic acid from the roots, and internally, where aluminum is detoxified by forming non-toxic aluminum-organic acid complexes that are stored in vacuoles. The ability to secrete oxalate to neutralize aluminum and to accumulate aluminum in vacuoles varies significantly between species and even between varieties within a species, resulting in different tolerance levels (Taunk *et al.*, 2023).

Highly tolerant genotypes to aluminum stress hold great potential as donor parents in plant breeding programs. Their superior traits can be combined with good agronomic traits and high yielding potential in a breeding program to develop high yielding and acidic adaptive cultivars. Consequently, the importance of these tolerant genotypes extends beyond solving agricultural problems, as they also drive innovation and advancement in the development of superior and sustainable plant varieties. STI-based ranking from controlled assays is an efficient pre-screen, but field validation is essential. Hydroponic or controlled screenings control the environment to observe genetic differences. However, soil chemistry, Al speciation, organic matter interactions, and microbiome effects in Ultisols can alter genotype performance under farmers' conditions (Dhandapani *et al.*, 2024). Thus, candidate genotypes identified here should be evaluated in representative acid-soil field trials.

The IC₇₅ estimates and model performance, ~1600 ppm AlCl₃·6H₂O is appropriate as a threshold concentration for discriminating aluminum-tolerant and sensitive cayenne pepper genotypes under deep water hydroponic conditions. Of the 41 genotypes evaluated, 'Tanggo' and 'CMK Lolay' exhibited high tolerance to aluminum stress, while 'PBC396', 'Laris', 'Perintis', 'Awe Aceh', 'Dumay', 'Iggo, Landung, Kawat, and Seulawah Aceh demonstrated moderate tolerance. Validation of tolerant genotypes in acid soil environments is required before integration into breeding pipelines. These findings suggest that these

genotypes possess varying degrees of resilience to aluminum toxicity, which is of valuable sources of Al tolerance controlling genes for future breeding programs.

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Conflicts of interest

The authors declare that there are no conflicts of interest regarding the publication of this paper.

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